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# Chemical Bonding: Hybridisation and Spatial Geometry: Construction

**Contents and concept route**

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**Sections+**

## Electron-Domain Counting and Molecular Shape Geometry

**Subject:** Chemistry

**Topic:** Chemical Bonding and Molecular Structure

**Concept:** Electron-Domain Counting and Molecular Shape Geometry

**Concept difficulty:** Medium concept

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## 1 Compare full local inventories, not only the common three sigma bonds.

Medium concept

Section 1 of 10

Next →

### Conditions for this construction

- Lewis structures
- VSEPR repulsion

#### Why this section matters for the question set

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### 1 Inventory bonds and central lone pairs separately.

**Why valid:** The givens supply three bonds at each centre but different lone-pair counts.

**Result:** BF<sub>3</sub>: 3+0=3 domains; NH<sub>3</sub>: 3+1=4.

### 2 The step the hard question turns on Apply the bounded introductory domain models.

**Why valid:** The local total, rather than bonds alone, selects the model arrangement.

**Result:** BF<sub>3</sub>: planar/sp<sup>2</sup>; NH<sub>3</sub>: tetrahedral domains/sp<sup>3</sup>-like.

**3 Name shape using bonded nuclei only.**

**Why valid:** The N lone-pair domain has no bonded nucleus.

**Result:** BF<sub>3</sub> is trigonal planar; NH<sub>3</sub> is trigonal pyramidal.

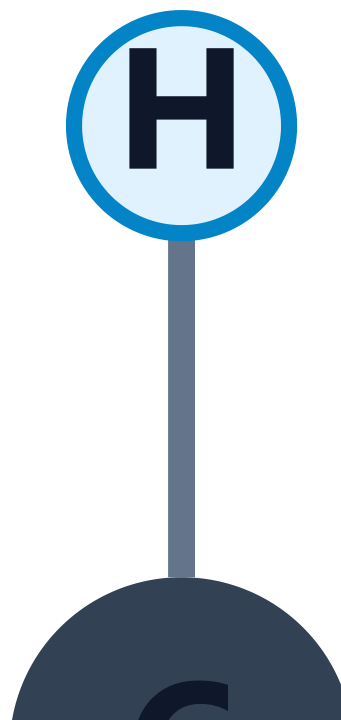
**4 Compare NH<sub>3</sub> with H<sub>2</sub>O.**

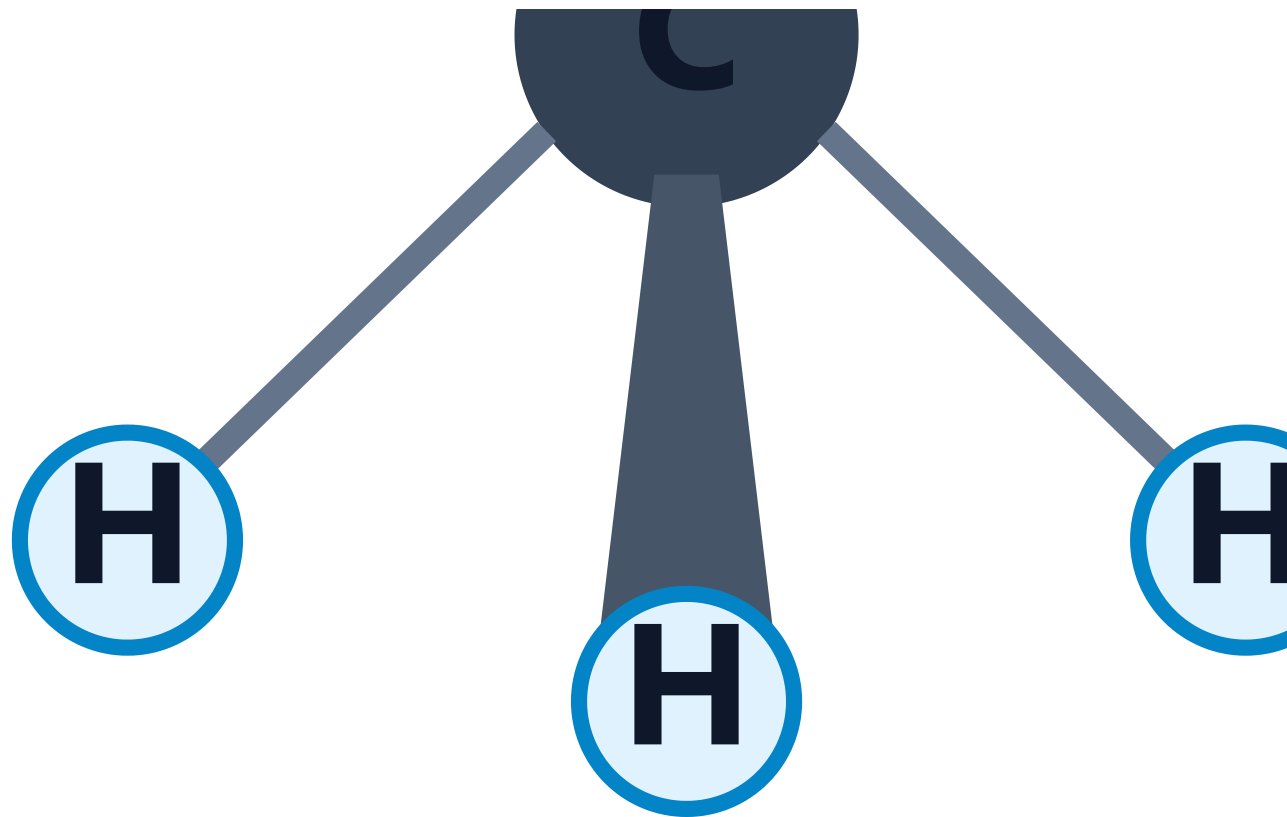
**Why valid:** Three-bond/one-pair and two-bond/two-pair inventories both total four.

**Result:** The same domain total can give different bonded-nucleus shapes.



**CH<sub>4</sub> (0 Lone Pairs)**





**Tetrahedral**

# Bond Angle: $109.5^\circ$

## Domain geom = Shape

CH<sub>4</sub> tetrahedral (0 LP)

Stage 1 of 3

Compare steric number 4 across CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O showing lone pair effect on shape.

### The key step

CH<sub>4</sub>, NH<sub>3</sub>, H<sub>2</sub>O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

### Worked explanation

Count local domains and bonded nuclei separately in BF<sub>3</sub>, NH<sub>3</sub> and H<sub>2</sub>O.

**Follow the operation, its justification and its result.**

1. A central lone pair adds a domain even though it adds no bonded nucleus.
2. Totals 3,4,4; bonded nuclei 3,3,2.
3. BF<sub>3</sub> has three B domains; NH<sub>3</sub> has three bonds plus one N lone pair. Introductory sp<sup>2</sup>/planar versus sp<sup>3</sup>-like domain/pyramidal shape follows the local model.

### Result

Totals 3,4,4; bonded nuclei 3,3,2.

### Check

Changed check: H<sub>2</sub>O and NH<sub>3</sub> have different numbers of bonds but the same four-domain total.

### Mistake clinic • diagnose and repair

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: H<sub>2</sub>O and NH<sub>3</sub> have different numbers of bonds but the same four-domain total.

#### 2 • APPLY

Compare a four-bond centre with a two-bond/two-lone-pair centre.

### 3 • CONNECT

A central lone pair adds a domain even though it adds no bonded nucleus.

## 2 Test whether three sigma bonds are sufficient for the proposed label.

Medium concept

Section 2 of 10

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Why this section matters for the question set

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### 1 Hold the claimed three-sigma condition fixed.

**Why valid:** Both supplied centres satisfy it, so an omitted lone-pair condition can test the implication.

**Result:** BF<sub>3</sub> and NH<sub>3</sub> both have three sigma bonds.

### 2 The step the hard question turns on Use nitrogen as the counterexample.

**Why valid:** NH<sub>3</sub> also has one central lone pair; its local total is four.

**Result:** Three sigma bonds alone do not imply three domains/sp<sup>2</sup>.

### 3 Replace the claim with a conditional local rule.

**Why valid:** Within localized introductory classification, bonds plus central lone pairs determine the domain

total.

**Result:** Use the full inventory: three domains -> sp<sup>2</sup> model; four -> sp<sup>3</sup>-like model.

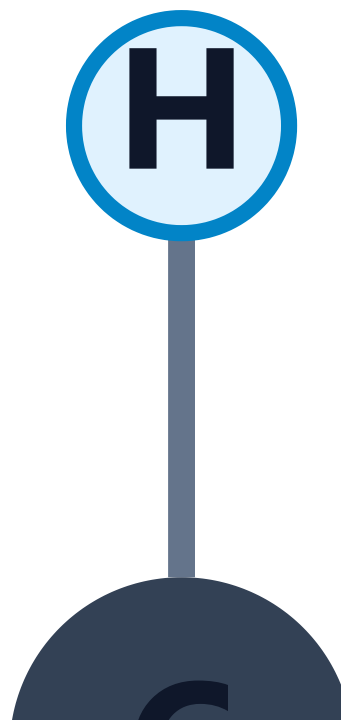
**4 Separate shape observations from a unique orbital basis.**

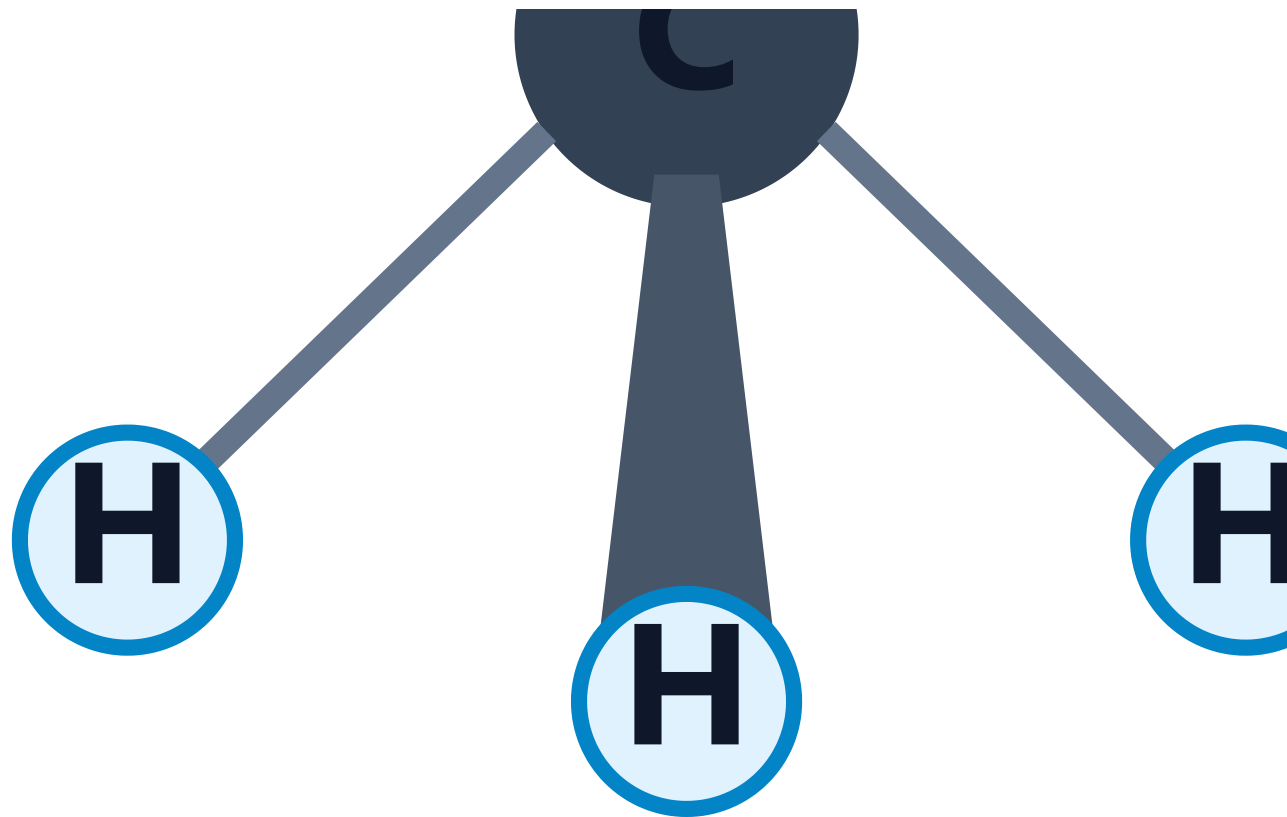
**Why valid:** Nuclear geometry constrains a description but does not measure hybrid coefficients.

**Result:** The counterexample rejects the bond-only rule without proving unique physical hybrid orbitals.



**CH<sub>4</sub> (0 Lone Pairs)**





**Tetrahedral**

# Bond Angle: $109.5^\circ$

## Domain geom = Shape

CH<sub>4</sub> tetrahedral (0 LP)

Stage 1 of 3

Compare steric number 4 across CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O showing lone pair effect on shape.

### Worked explanation

Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.

**Follow the operation, its justification and its result.**

1. A counterexample must hold the claimed condition while changing an omitted relevant condition.

2. No: four versus three. Keeping the answer identical despite the changed lone-pair condition suggests the misconception.
3. NH3 falsifies three-sigma-implies-sp2: its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.

### Result

No: four versus three. Keeping the answer identical despite the changed lone-pair condition suggests the misconception.

### Check

Changed check: BF3 and NH3 share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

### Mistake clinic • diagnose and repair



#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed check: BF3 and NH3 share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

## 2 • APPLY

Repair the rule with an explicit applicability condition.

## 3 • CONNECT

A counterexample must hold the claimed condition while changing an omitted relevant condition.

### 3 Separate number of bonding directions from bond identity or strength.

Medium concept

Section 3 of 10

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Why this section matters for the question set

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#### 1 Recount carbon neighbours after H-to-F substitution.

**Why valid:** CH3F has three H and one F directly attached to C.

**Result:** Four carbon bonding directions.

#### 2 The step the hard question turns on Apply the same local four-domain allocation.

**Why valid:** Changing substituent identity does not itself create a new neighbour or lone pair at C.

**Result:** Introductory  $sp^3$  carbon model.

**3 State what the count leaves open.**

**Why valid:** Bond lengths, polarity and exact angles require information beyond neighbour counting.

**Result:** No equality of C-H and C-F properties is inferred.

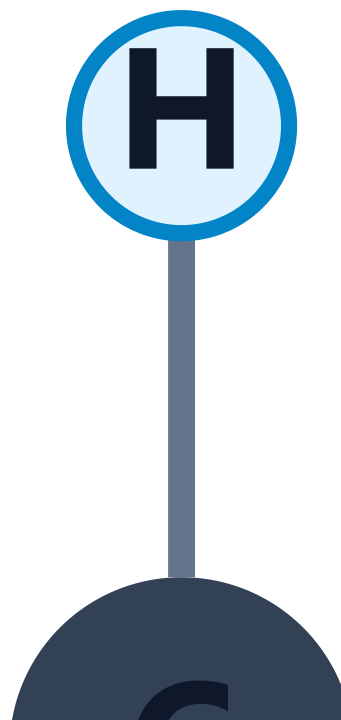
**4 Test CH<sub>2</sub>F<sub>2</sub> as a changed inventory.**

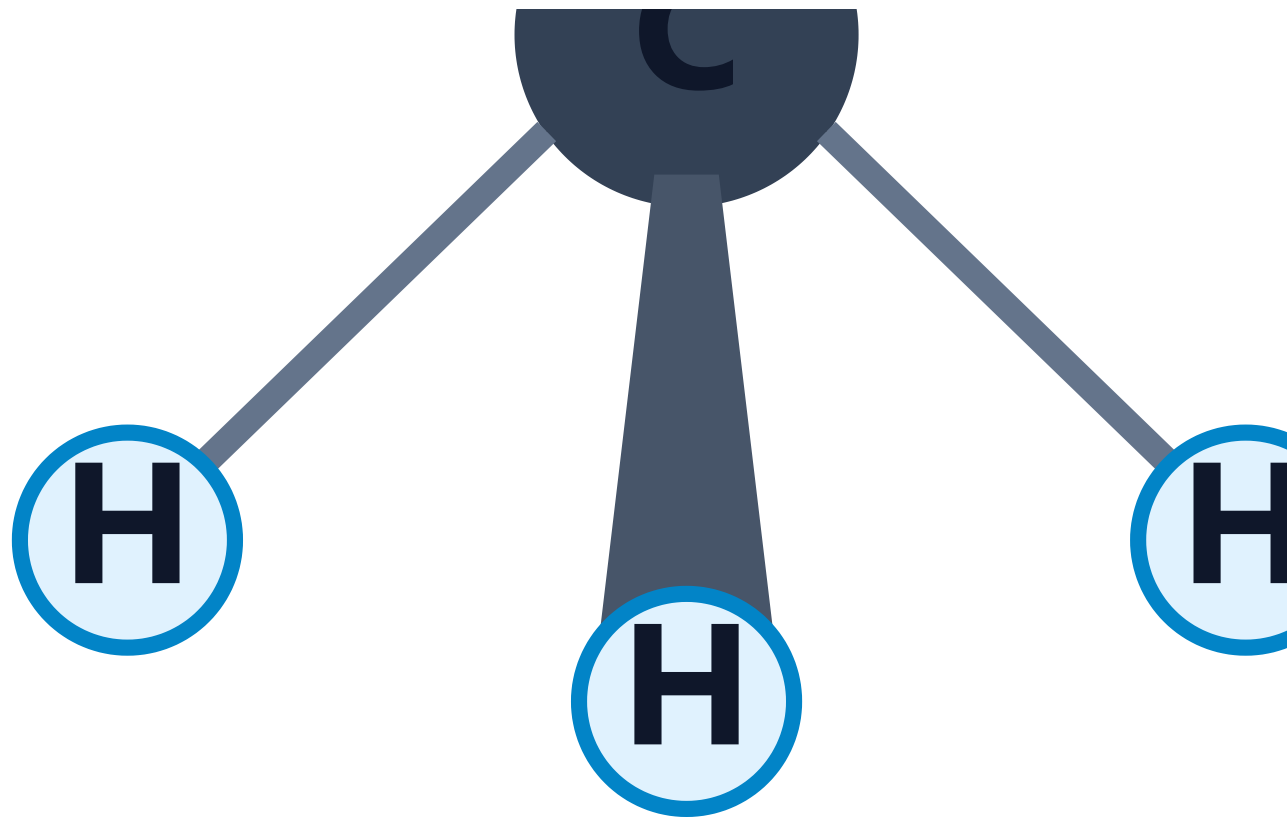
**Why valid:** Two H and two F are still four neighbours.

**Result:** Four domains remain; electronegativity is not an extra domain.



**CH<sub>4</sub> (0 Lone Pairs)**





**Tetrahedral**

# Bond Angle: $109.5^\circ$

## Domain geom = Shape

CH<sub>4</sub> tetrahedral (0 LP)

Stage 1 of 3

Compare steric number 4 across CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O showing lone pair effect on shape.

### Worked explanation

Count carbon domains in CH<sub>4</sub>, CH<sub>3</sub>F and CH<sub>2</sub>F<sub>2</sub>. Does each F add an extra domain beyond its C-F direction?

**Follow the operation, its justification and its result.**

1. Replacing a substituent changes chemical environment without automatically changing connectivity count.

2. All four; no.
3.  $\text{CH}_3\text{F}$  carbon has four bonding directions and introductory  $\text{sp}^3$  accounting. The count alone establishes neither bond lengths nor their detailed causes.

### Result

All four; no.

### Check

Changed check:  $\text{CH}_2\text{F}_2$  still has four central bonding directions; electronegativity is not an extra domain.

### Mistake clinic • diagnose and repair



#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed check:  $\text{CH}_2\text{F}_2$  still has four central bonding directions; electronegativity is not an extra domain.

##### 2 • APPLY

Classify carbon in  $\text{CHCl}_3$  without assigning an exact bond angle.

##### 3 • CONNECT

Replacing a substituent changes chemical environment without automatically changing connectivity count.

## 4 Count local domains and bonded nuclei at two different centres.

Medium concept

Section 4 of 10

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Why this section matters for the question set

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### 1 Make atom-local inventories in CH<sub>3</sub>NH<sub>2</sub>.

**Why valid:** C has four single bonds; N has three single bonds and one lone pair.

**Result:** C: 4+0; N: 3+1; both total four domains.

### 2 The step the hard question turns on Assign the localized introductory domain models.

**Why valid:** Each four-domain centre can use sp<sup>3</sup>-like accounting under the stated local model.

**Result:** Four model directions at each centre.

### 3 Count nuclei for the separate shape descriptions.

**Why valid:** C has four bonded neighbours while N has three; a lone pair is not an atomic vertex.

**Result:** Tetrahedral C nuclear arrangement; pyramidal N arrangement.

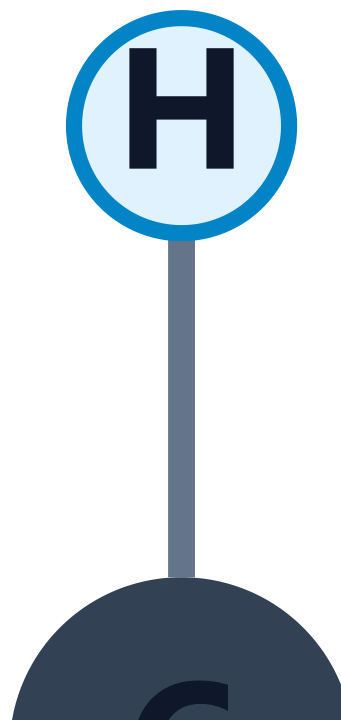
**4 Replace an N-H bond with an N-CH<sub>3</sub> bond in a neutral amine comparison.**

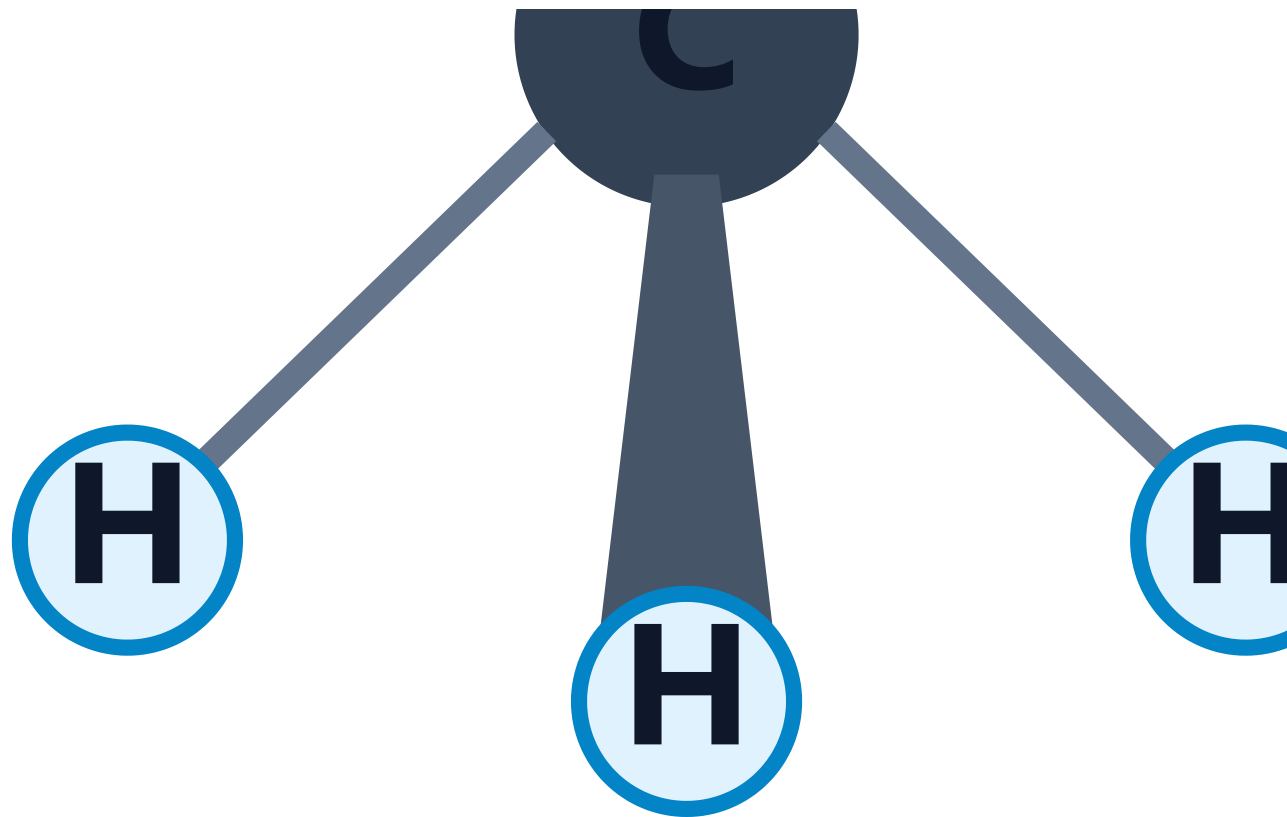
**Why valid:** The N inventory stays three bonds plus one lone pair.

**Result:** Local N count and pyramidal introductory description persist; exact angle is unresolved.



**CH<sub>4</sub> (0 Lone Pairs)**





**Tetrahedral**

# Bond Angle: $109.5^\circ$

## Domain geom = Shape

CH<sub>4</sub> tetrahedral (0 LP)

Stage 1 of 3

Compare steric number 4 across CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O showing lone pair effect on shape.

### Worked explanation

Compare local N inventory with carbon inventory in CH<sub>3</sub>NH<sub>2</sub>. Which centre includes a lone-pair direction?

**Follow the operation, its justification and its result.**

1. A lone pair contributes to electron-domain geometry but is excluded from molecular-shape naming.

2. N does; C does not. Using one atom-global shape for both reveals locality confusion.
3. Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.

### Result

N does; C does not. Using one atom-global shape for both reveals locality confusion.

### Check

Changed check: replacing N-H with N-CH<sub>3</sub> preserves the local three-bond-plus-lone-pair N inventory in the neutral localized amine model.

### Mistake clinic • diagnose and repair

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: replacing N-H with N-CH<sub>3</sub> preserves the local three-bond-plus-lone-pair N inventory in the neutral localized amine model.

#### 2 • APPLY

Describe a neutral secondary amine N without predicting an exact experimental angle.

### 3 • CONNECT

A lone pair contributes to electron-domain geometry but is excluded from molecular-shape naming.

## 5 Test whether domain geometry and nuclear shape are interchangeable.

Medium concept

Section 5 of 10

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Why this section matters for the question set

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### 1 Hold the four-domain total fixed across three cases.

**Why valid:** CH<sub>4</sub>, NH<sub>3</sub> and H<sub>2</sub>O have respective bond/pair inventories 4/0, 3/1 and 2/2.

**Result:** All have four introductory domains.

### 2 The step the hard question turns on Use all domains for domain geometry.

**Why valid:** Bonding and lone-pair entries occupy the domain-model directions.

**Result:** Tetrahedral electron-domain arrangement in all three.

### 3 Use bonded nuclei for molecular shape.

**Why valid:** Removing LP entries from the shape name changes the number of atomic vertices.

**Result:** CH<sub>4</sub> tetrahedral; NH<sub>3</sub> pyramidal; H<sub>2</sub>O bent.

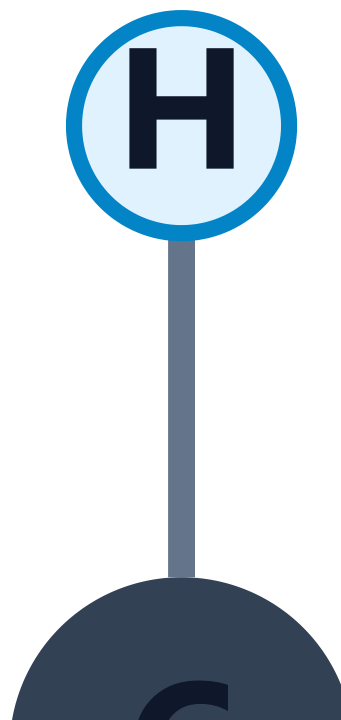
**4 State the condition for the two names to coincide.**

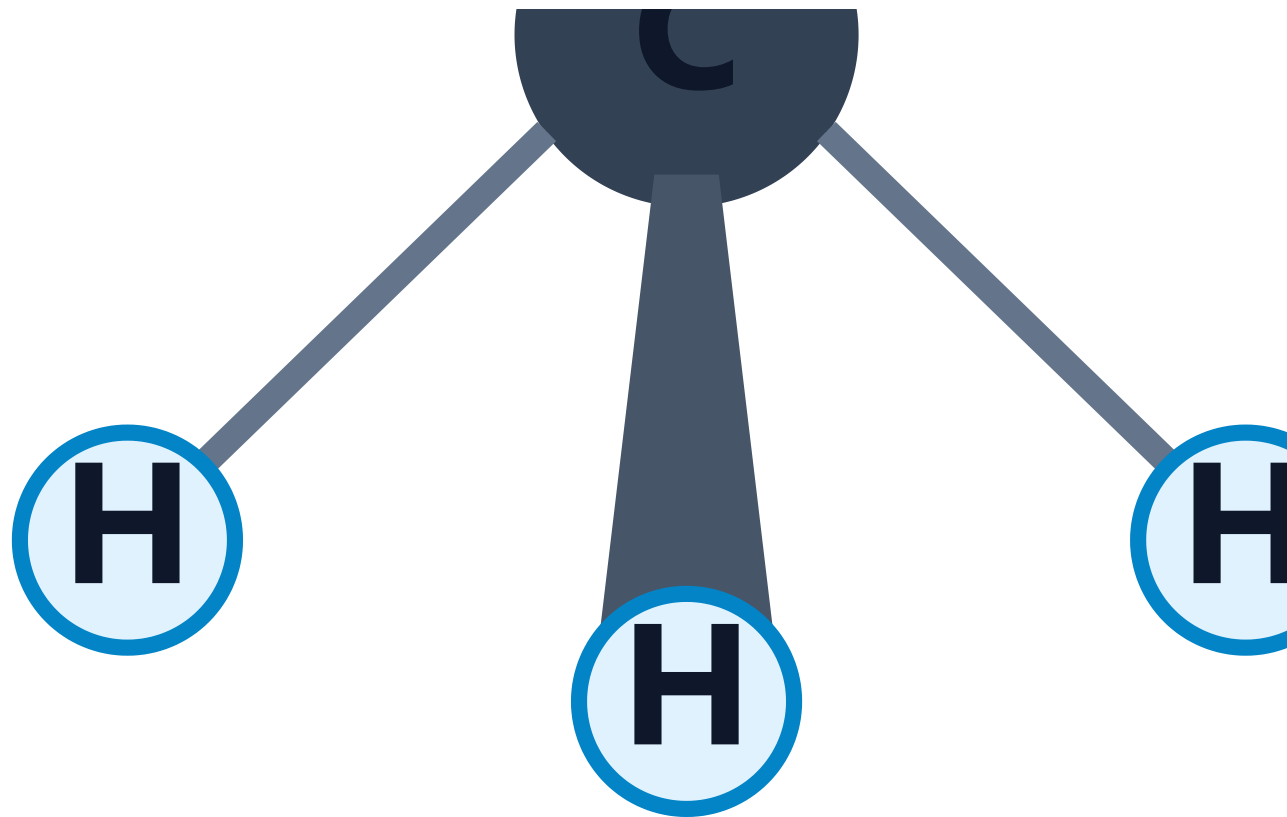
**Why valid:** When there is no central lone pair every domain direction terminates at a bonded nucleus.

**Result:** The student claim holds for the no-lone-pair case, not universally.



**CH<sub>4</sub> (0 Lone Pairs)**





**Tetrahedral**

# Bond Angle: $109.5^\circ$

## Domain geom = Shape

CH<sub>4</sub> tetrahedral (0 LP)

Stage 1 of 3

Compare steric number 4 across CH<sub>4</sub>, NH<sub>3</sub>, and H<sub>2</sub>O showing lone pair effect on shape.

### Worked explanation

For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

**Follow the operation, its justification and its result.**

1. A lone-pair domain has no bonded nucleus to include in molecular-shape naming.

2. Tetrahedral, pyramidal, bent; omitted slots are lone pairs. Repeating tetrahedral for all atom-only shapes exposes the conflation.
3. CH<sub>4</sub>, NH<sub>3</sub>, H<sub>2</sub>O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

### Result

Tetrahedral, pyramidal, bent; omitted slots are lone pairs. Repeating tetrahedral for all atom-only shapes exposes the conflation.

### Check

Changed check: a four-bond/no-lone-pair centre cannot be assigned the same atom-only shape as a two-bond/two-lone-pair centre.

#### Mistake clinic • diagnose and repair



#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed check: a four-bond/no-lone-pair centre cannot be assigned the same atom-only shape as a two-bond/two-lone-pair centre.

## 2 • APPLY

Sketch the domain and atom-only inventories separately for H<sub>2</sub>O.

## 3 • CONNECT

A lone-pair domain has no bonded nucleus to include in molecular-shape naming.

### Question-specific diagnosis and repair



### Now you do one

Given OF<sub>2</sub> with two O-F bonds and two O lone pairs, classify the domain arrangement and bonded-nucleus shape at O. Explain why terminal-F lone pairs are excluded.

### Practice this concept in Core2

Q1

Q5

Q6

Q8

Q10

**Sections+**

## Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

**Subject:** Chemistry**Topic:** Chemical Bonding and Molecular Structure**Concept:** Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination**Concept difficulty:** Medium concept**1**

**Conserve carbon orbitals while separating domains from sigma/pi components.**

**Medium concept**

Section 6 of 10

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### Conditions for this construction

- Lewis structures
- VSEPR electron-domain counting

### Why this section matters for the question set

+

**1 Count C neighbours in HCHO.**

**Why valid:** Two H and one O are three connections; bond multiplicity is a separate inventory.

**Result:** Three domains; no C lone pair.

**2 The step the hard question turns on Allocate the four carbon valence functions.**

**Why valid:** One 2s plus two 2p form three  $sp^2$  hybrids, leaving the third 2p unused in that mixing.

**Result:** Three sigma-direction hybrids plus one residual p.

**3 Resolve the C=O bond components.**

**Why valid:** A localized double-bond account has one axial sigma and one lateral pi component.

**Result:** C=O: one sigma + one pi; residual C p supplies the pi role.

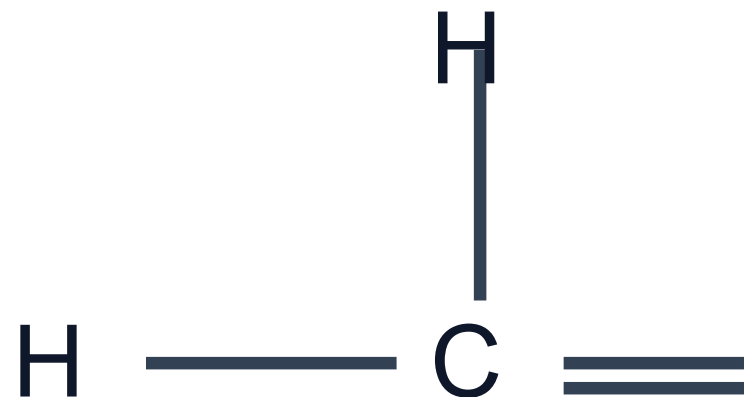
**4 Close the basis ledger using an sp comparison.**

**Why valid:** sp gives two hybrids and two residual p functions from the same four inputs.

**Result:** Both allocations total four; no p function is spent twice.



H<sub>2</sub>C=O connectivity



H<sub>2</sub>C=O connectivity

# Connectivity layout only; page angles are not measured

H<sub>2</sub>C=O connectivity

Stage 1 of 2

Allocate the same carbon valence basis to the requested sigma and pi jobs.

## The key step

CO<sub>2</sub> carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful for accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.

## Worked explanation

Inventory sp<sup>2</sup> carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?

**Follow the operation, its justification and its result.**

1. Three sigma directions do not consume the remaining perpendicular p orbital.
2. No double use. Totals remain four in both cases.
3. HCHO carbon has three-domain  $sp^2$  accounting, three  $sp^2$  sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.

### Result

No double use. Totals remain four in both cases.

### Check

Changed check: for an sp carbon, two hybrids plus two residual p orbitals still total four basis functions.

### Mistake clinic • diagnose and repair

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: for an sp carbon, two hybrids plus two residual p orbitals still total four basis functions.

#### 2 • APPLY

Inventory HCN carbon without copying the formaldehyde count.

### 3 • CONNECT

Three sigma directions do not consume the remaining perpendicular p orbital.

## 2 Construct the relation between the molecular plane and the pi orbitals.

Medium concept

Section 7 of 10

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Next →

Why this section matters for the question set

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### 1 Build the two local carbon sigma frames.

**Why valid:** Each ethene carbon has three sigma directions and one remaining p function in  $sp^2$  accounting.

**Result:** Two in-plane three-direction frames; one perpendicular p at each C.

### 2 The step the hard question turns on Align the p functions relative to the molecular plane.

**Why valid:** A residual p is perpendicular to its carbon sigma plane.

**Result:** Parallel perpendicular p directions in planar ethene.

### 3 Draw both axial and lateral overlap.

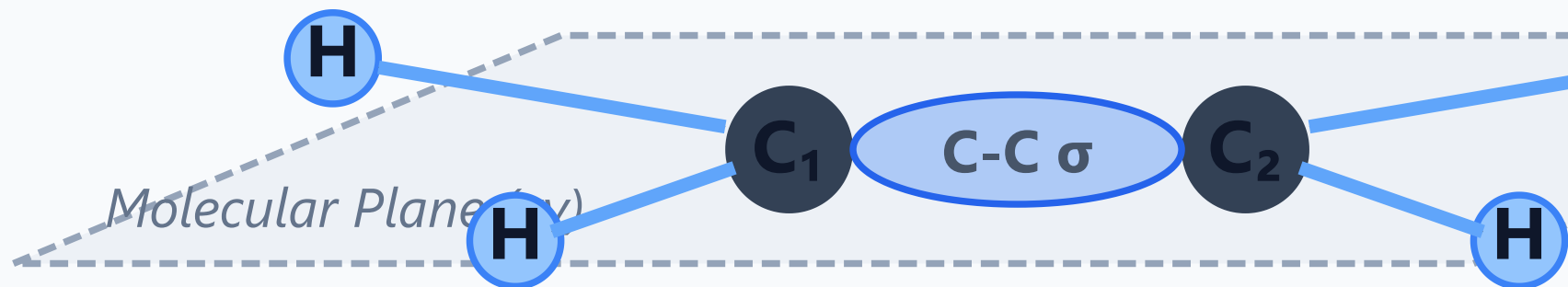
**Why valid:** The C-C sigma component is axial; lateral p overlap lies above and below the plane.

**Result:** A sigma framework plus a pi component with a node in the molecular plane.

**4 Compare rigid rotation with relative CH<sub>2</sub>-group rotation.**

**Why valid:** Only the latter changes relative p alignment.

**Result:** Whole-picture rotation preserves overlap geometry; relative twisting can reduce pi alignment.



In-plane  $\sigma$ -framework: 5  $\sigma$  bonds (1 C-C, 4 C-H) lie in the n

In-plane sp<sup>2</sup> sigma framework

Stage 1 of 2

Model planar sigma backbone and perpendicular pi overlap in ethene.

### Worked explanation

Rotate the whole ethene picture, then rotate only one CH<sub>2</sub> group. Which operation changes relative p alignment?

**Follow the operation, its justification and its result.**

1. Three sigma directions lie in each sp<sup>2</sup> carbon plane; residual p is perpendicular to that plane.
2. Only the relative-group rotation changes alignment. A whole-page-orientation answer exposes a representation error.
3. Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.

### Result

Only the relative-group rotation changes alignment. A whole-page-orientation answer exposes a representation error.

### Check

Changed check: rotate one CH<sub>2</sub> group while preserving C-C connectivity; ask which overlap is sensitive to the relative orientation.

### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: rotate one CH<sub>2</sub> group while preserving C-C connectivity; ask which overlap is sensitive to the relative orientation.

#### 2 • APPLY

Sketch a qualitative twisted ethene comparison without predicting a numerical barrier.

#### 3 • CONNECT

Three sigma directions lie in each sp<sup>2</sup> carbon plane; residual p is perpendicular to that plane.

3

**The central carbon inventory must support two directions and two pi components.**

Medium concept

Section 8 of 10

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Next →

**1 Count two O neighbours at CO<sub>2</sub> carbon.**

**Why valid:** Each C=O multiple bond contributes one local direction.

**Result:** Two carbon domains; linear introductory framework.

**2 The step the hard question turns on Allocate carbon orbitals without duplication.**

**Why valid:** One 2s plus one 2p yields two sp hybrids; two p directions remain perpendicular.

**Result:** Two sigma directions and two residual C p functions.

**3 Account for each localized C=O component.**

**Why valid:** Two double bonds require two sigma and two pi components in this bookkeeping model.

**Result:** Two sigma + two pi; all four carbon functions have declared roles.

**4 Bound the inference about terminal oxygen.**

**Why valid:** A localized basis drawing is one representation of the electronic description; carbon counting does not uniquely measure an oxygen basis.

**Result:** Linear carbon accounting is consistent; a unique observed O orbital-plane picture is not established.

# Stage 1: Lewis Domain Co

$:\ddot{\text{O}} = \text{C} = \ddot{\text{O}}:$  → 2 Electron Do

Double bond = 1 domain; Linea

Lewis domain count

Stage 1 of 4

Model linear sp hybridisation and orthogonal unhybridised p orbital overlap in CO<sub>2</sub>.

### **Worked explanation**

Give carbon orbital inventory for CO<sub>2</sub> and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?

### **Follow the operation, its justification and its result.**

1. sp accounting leaves two mutually perpendicular p directions.
2. Both carbon inventories close at four. No unique O basis follows from the drawing alone.
3. CO<sub>2</sub> carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.

### **Result**

Both carbon inventories close at four. No unique O basis follows from the drawing alone.

### **Check**

Changed check: HCN carbon also has two sigma directions and two residual p directions although its neighbours differ.

**Mistake clinic • diagnose and repair**

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### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: HCN carbon also has two sigma directions and two residual p directions although its neighbours differ.

#### 2 • APPLY

State what CO<sub>2</sub> geometry supports and what it cannot uniquely prove.

#### 3 • CONNECT

sp accounting leaves two mutually perpendicular p directions.

## 4 Keep local inventories separate from the triple-bond component inventory.

Medium concept

Section 9 of 10

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Next →

**Why this section matters for the question set**

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**1 Inventory carbon and nitrogen separately in HCN.**

**Why valid:** C has H and N neighbours; N has one C neighbour plus one lone pair.

**Result:** C: two domains; N: two domains.

**2 The step the hard question turns on Allocate each local four-function valence basis.**

**Why valid:** Two sp hybrids leave two perpendicular p functions at each second-period centre.

**Result:** Introductory sp at C and N.

**3 Count triple-bond components without adding local directions.**

**Why valid:** Axial overlap gives one sigma; the two residual p pairs give two pi components.

**Result:** C-N: one sigma + two pi; C-H adds one sigma.

**4 Close the ten-valence-electron inventory.**

**Why valid:** H+C+N contributes  $1+4+5=10$ ; four shared pairs and one N lone pair use five pairs.

**Result:** Eight bonding electrons + two lone-pair electrons = ten.



# Supplied HCN Lewis connectivity



HCN connectivity; : is the 1

# Connectivity layout only; page angles are not measured

Supplied HCN Lewis connectivity

Stage 1 of 2

Make separate local inventories at C and N; bond multiplicity is a separate ledger.

## Worked explanation

Compare domain counts at C and N with the total number of shared pairs in HCN.

**Follow the operation, its justification and its result.**

1. One bonded neighbour supplies one direction even when three bond components are present.
2. Both local domain counts are two; four shared pairs plus N lone pair close the ten-electron ledger.

3. HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.

### Result

Both local domain counts are two; four shared pairs plus N lone pair close the ten-electron ledger.

### Check

Changed check: total valence electrons HCN =  $1+4+5 = 10$ ; the three shared pairs and N lone pair plus C-H pair use all ten.

### Mistake clinic • diagnose and repair

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#### 1-2-3 QUICK CHECK

##### 1 • CHECK

Changed check: total valence electrons HCN =  $1+4+5 = 10$ ; the three shared pairs and N lone pair plus C-H pair use all ten.

##### 2 • APPLY

Compare HCN and CO<sub>2</sub> local carbon inventories.

##### 3 • CONNECT

One bonded neighbour supplies one direction even when three bond components are present.

## 5 Avoid spending a carbon basis function twice.

Medium concept

Section 10 of 10

← Previous

Why this section matters for the question set

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### 1 Start from four carbon valence-basis functions.

**Why valid:** The second-period basis is one 2s plus three 2p.

**Result:** Four inputs; three sigma directions and one pi role required.

### 2 The step the hard question turns on Mix only the functions assigned to sigma directions.

**Why valid:** One s plus two p makes three in-plane  $sp^2$  hybrids.

**Result:** Three hybrids; one p remains outside the mixing.

### 3 Assign the residual p to the pi role.

**Why valid:** It is perpendicular to the sigma plane and can overlap laterally with a suitable neighbour.

**Result:** Three sigma hybrids + one residual perpendicular p.

**4 Check against sp and sp<sup>3</sup> allocations.**

**Why valid:** Those alternatives account for two hybrids plus two p, or four hybrids plus zero p.

**Result:** All totals are four; no function is simultaneously a hybrid input and an extra residual p.



# Supplied local carbon jobs

Three sigma directions; no lone pair

One pi participation

Available valence basis: one 2s and three

# Connectivity layout only; page angles are not measured

Supplied local carbon jobs

Stage 1 of 2

Allocate each basis function once. The completed orbital arrangement is your response.

## Worked explanation

Build an allocation ledger for sp, sp<sup>2</sup>, sp<sup>3</sup> and flag any duplicate-use entry.

**Follow the operation, its justification and its result.**

1. For second-period carbon, one 2s and three 2p functions form a four-dimensional valence basis.
2. All totals four; residual p counts 2,1,0.

3.  $sp^2$  accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

### Result

All totals four; residual p counts 2,1,0.

### Check

Changed check: sp accounting gives two hybrids plus two residual p;  $sp^3$  gives four hybrids plus zero residual p.

### Mistake clinic • diagnose and repair

+

### 1-2-3 QUICK CHECK

#### 1 • CHECK

Changed check: sp accounting gives two hybrids plus two residual p;  $sp^3$  gives four hybrids plus zero residual p.

#### 2 • APPLY

Allocate the  $sp^2$  basis without memorising an orbital picture.

#### 3 • CONNECT

For second-period carbon, one 2s and three 2p functions form a four-dimensional valence basis.

### Question-specific diagnosis and repair

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#### Now you do one

For  $\text{H}-\text{C}\equiv\text{C}-\text{H}$ , inventory domains and residual p functions at each C, then count sigma and pi components in the entire molecule. Close the valence-electron ledger.

#### Practice this concept in Core2

Q2

Q3

Q4

Q7

Q9